The documentation and process conversion measures necessary to comply with this revision shall be completed by 1 January 1998

INCH-POUND

MIL-PRF-19500/452D
1 October 1997
SUPERSEDING
MIL-S-19500/452C
2 August 1993

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, LOW LEVEL VOLTAGE-REFERENCE,
TYPES 1N4565A-1 THROUGH 1N4584A-1, AND 1N4565AUR-1 THROUGH 1N4584AUR-1
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC; RADIATION HARDENED (TOTAL DOSE ONLY)
TYPES JANTXVM, D, L, R, F, G, H; JANSM, D, L, R, F, G, H;
JANHCM, D, L, R, F, G, H; AND JANKCM, D, L, R, F, G, H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for 6.4 volts ±5 percent, silicon, low bias current, voltage-reference diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-PRF-19500, and two levels of product assurance for each unencapsulated device type die. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type, and two levels of product assurance for each unencapsulated device type die as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (DO-7 and DO-35), figure 2 (DO-213AA), figure 3 (JANHCA and JANKCA), and figure 4 (JANHCB and JANKCB).

1.3 Unless otherwise specified, maximum ratings (T_A = +25°C).

P_T	T_{STG} and T_{op}	I_{ZM} 1/	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-65 to +175	70	3.33

1/ To guarantee voltage temperature stalility, it is necessary to maintain the proper Iz as specified in paragraph 1.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 0.5.mA dc	V_Z I_Z = 0.5 mA dc		I_R V_R = 3.0 V
			Min	Max	
	mV dc	ohms	Volts	Volts	µA
1N4565A-1, 1N4565AUR-1	0.100	200	6.08	6.72	2.0
1N4566A-1, 1N4566AUR-1	0.050	"	"	"	"
1N4567A-1, 1N4567AUR-1	0.020	"	"	"	"
1N4568A-1, 1N4568AUR-1	0.010	"	"	"	"
1N4569A-1, 1N4569AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 1.0 mA dc	V_Z I_Z = 1.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	mV dc	ohms	Volts	Volts	µA
1N4570A-1, 1N4570AUR-1	0.100	100	6.08	6.72	2.0
1N4571A-1, 1N4571AUR-1	0.050	"	"	"	"
1N4572A-1, 1N4572AUR-1	0.020	"	"	"	"
1N4573A-1, 1N4573AUR-1	0.010	"	"	"	"
1N4574A-1, 1N4574AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 2.0 mA dc	V_Z I_Z = 2.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	mV dc	ohms	Volts	Volts	µA
1N4575A-1, 1N4575AUR-1	0.100	50	6.08	6.72	2.0
1N4576A-1, 1N4576AUR-1	0.050	"	"	"	"
1N4577A-1, 1N4577AUR-1	0.020	"	"	"	"
1N4578A-1, 1N4578AUR-1	0.010	"	"	"	"
1N4579A-1, 1N4579AUR-1	0.005	"	"	"	"

Type 1/	ΔV_Z (voltage-temperature stability)	Z_Z I_Z = 4.0 mA dc	V_Z I_Z = 4.0 mA dc		I_R V_R = 3.0 V
			Min	Max	
	mV dc	ohms	Volts	Volts	µA
1N4580A-1, 1N4580AUR-1	0.100	25	6.08	6.72	2.0
1N4581A-1, 1N4581AUR-1	0.050	"	"	"	"
1N4582A-1, 1N4582AUR-1	0.020	"	"	"	"
1N4583A-1, 1N4583AUR-1	0.010	"	"	"	"
1N4584A-1, 1N4584AUR-1	0.005	"	"	"	"

1/ Electrical characteristics and test conditions for " A-1, and AUR-1" devices are identical.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item performance requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be specified in MIL-PRF-19500 and figures 1, 2, 3, and 4 herein.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein.

3.4.2 Diode construction. These devices shall be constructed in a manner and using materials which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

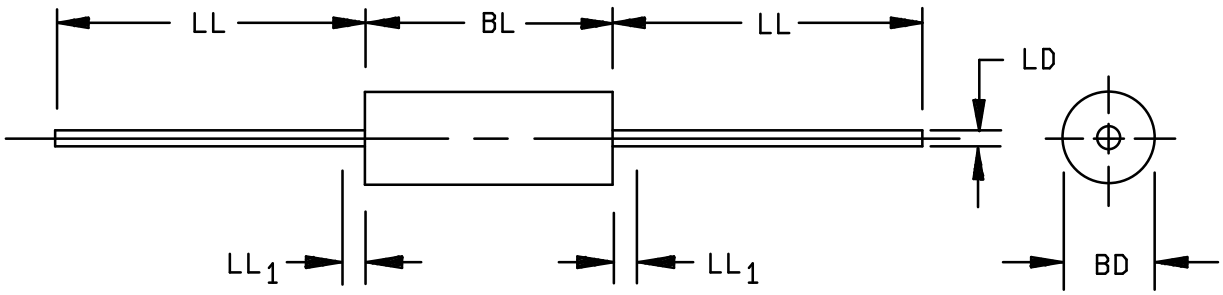
3.4.3 Dash-one construction. Shall be as specified in MIL-PRF-19500.

3.4.4 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-PRF-19500.

3.4.5 JANHC and JANKC construction. JANHC and JANKC construction may differ in die size and bonding pad layout provided the manufacturing technology is identical (example: Diffused junction, alloy junction).

3.5 Marking. Devices shall be marked as specified in MIL-PRF-19500.

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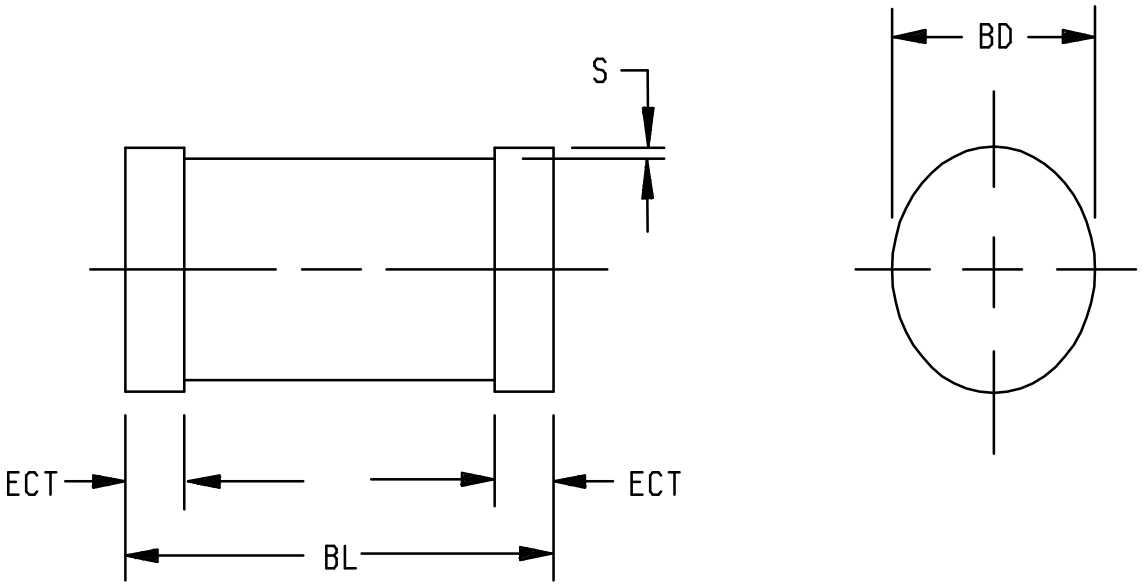


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.060	.107	1.52	2.72	3
BL	.120	.300	3.05	7.62	3
LD	.018	.023	0.46	0.58	
LL	1.000	1.500	25.40	38.10	
LL_1	---	0.50	---	1.27	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within BD and length BL. Heat slugs, if any, shall be included within this cylinder but shall not be subject to minimum limit of BD.
4. Within this zone, lead diameter may vary to allow for lead finishes and irregularities, other than heat slugs.

FIGURE 1. Physical dimensions 1N4565A-1 THROUGH 1N4584A-1,(DO-35 and DO-7).

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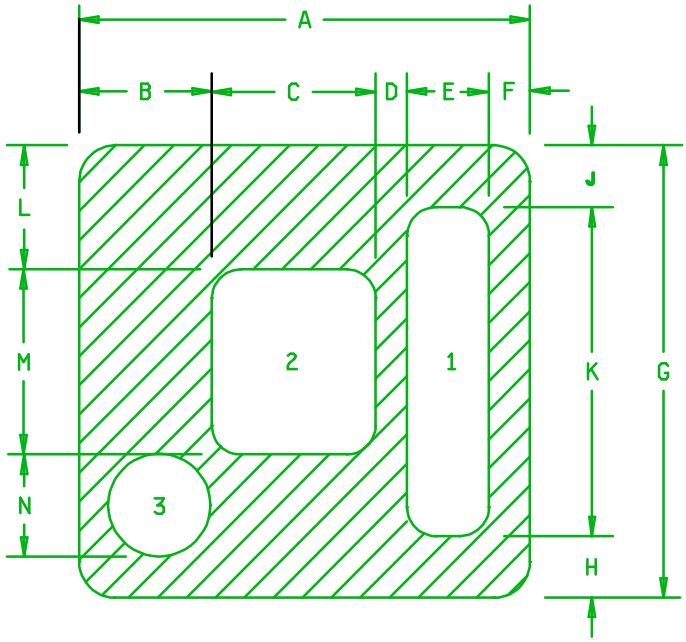
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BD	.063	.067	1.60	1.70
ECT	.016	.022	0.41	0.56
BL	.130	.146	3.30	3.71
S	.001 Min		0.03 Min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions 1N4565AUR-1 THROUGH 1N4584AUR-1 (DO-213AA).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0280	.0320	.711	.813
B	.0080	.0100	.203	.254
C	.0104	.0106	.264	.269
D	0019	.0021	.048	.053
E	.0054	.0056	.137	.142
F	.0020	.0040	.050	.102
G	.0280	.0320	.711	.813
H	.0030	.0050	.076	.127
J	.0030	.0050	.076	.127
K	.0209	.0211	.531	.536
L	.0080	.0100	.203	.254
M	.0104	.0106	.264	.269
N	.0059	.0061	.150	.155



BACKSIDE MUST BE ELECTRICALLY ISOLATED

DESIGN DATA

Metallization:

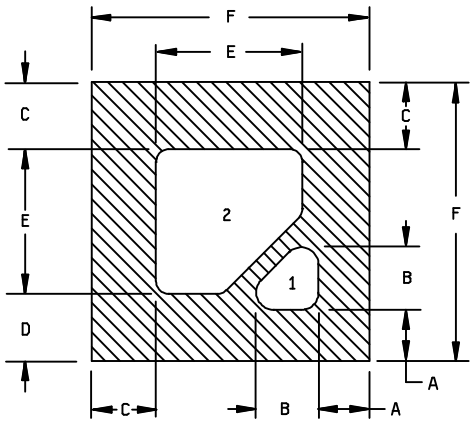
Top:	1 (Cathode)	Al	Circuit layout data:
	2 (Anode)	Al	For zener operation, cathode must be
	3 (Test pad)	Al	operated positive with respect to anode.
Back	Au		

Al thickness	25,000Å minimum
Gold thickness	4,000Å minimum
Chip thickness	0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches unless otherwise indicated.
2. Metric equivalents are given for general information only.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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BACKSIDE MUST BE ELECTRICALLY ISOLATED

Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.0035	.0065	.088	.165
B	.0050	.0080	.127	.200
C	.0050	.0065	.127	.165
D	.0050	.0065	.127	.165
E	.0150	.0165	.381	.419
F	.0260	.0290	.660	.737

DESIGN DATA

Metallization:

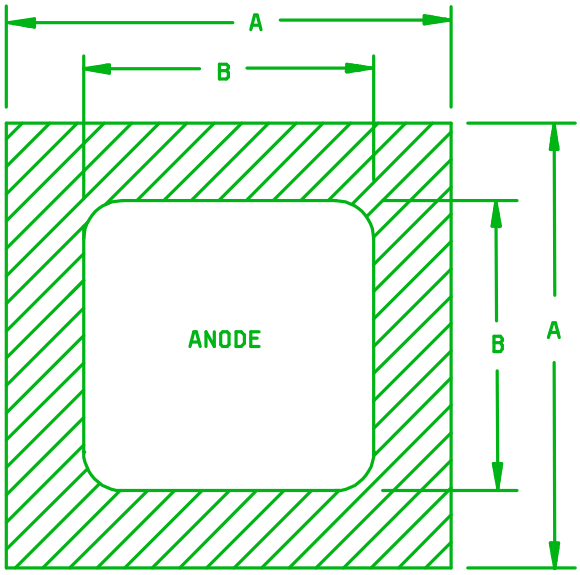
Circuit layout data: Top: (Anode) . . . Al
For zener operation, cathode must be
operated positive with respect to anode. Back . . . . . Au

Al thickness . . . . . 40,000Å minimum
Gold thickness . . . . . 5,000Å minimum
Chip thickness . . . . . 0.010 inch (0.254 mm) ±0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.

FIGURE 4. JANHC and JANKC (B-version) die dimensions.

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Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
A	.024	.037	0.61	0.94
B	.014	.029	0.36	0.74

DESIGN DATA

Metallization:

Circuit layout data
For zener operation, cathode must be operated positive with respect anode.

Top:	(Anode)	Al
Back		Au
Al thickness		40,000Å minimum
Gold thickness		5,000Å minimum
Chip thickness		0.010 inch (0.254 mm) ± 0.002 inch (0.051 mm).

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Requirements in accordance with MIL-PRF-19500 (appendix G) are performed in a TO-5 package (see 6.5).

FIGURE 5. JANHC and JANKC (C-version) die dimensions.

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3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.3).

4.1.1 Sampling inspection. Sampling inspection shall be in accordance with MIL-PRF-19500 and as specified herein, except that lot accumulation period shall be 6 months in lieu of 6 weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500 and paragraph 4.4.5 herein.

4.2.2 JANHC and JANKC devices. Qualification for JANHC and JANKC devices shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
11	V_Z, Z_Z, and I_R	V_Z, Z_Z, and I_R
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading $\Delta V_Z \leq \pm 0.005$ V dc from initial value at T_A = +25°C ±2°C	Subgroup 2 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading. $\Delta V_Z \leq \pm 0.005$ V dc from initial value at T_A = +25°C ±2°C

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

1N4565 through 1N4569: I_Z = 0.5 mA dc ±0.05 mA dc, T_A = +150 +5°C, -0°C.
1N4570 through 1N4574: I_Z = 1.0 mA dc ±0.1 mA dc, T_A = +150 +5°C, -0°C.
1N4575 through 1N4579: I_Z = 2.0 mA dc ±0.2 mA dc, T_A = +150 +5°C, -0°C.
1N4580 through 1N4584: I_Z = 4.0 mA dc ±0.4 mA dc, T_A = +150 +5°C, -0°C.

To better utilize burn-in equipment, higher values of I_Z shall be permitted provided the junction temperature does not exceed +175°C.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with appendix E, table V of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, tables VIa (JANS) and VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, appendix E, table VIa (JANS) of MIL-PRF-19500. For purposes of JANS inspection, a single device type shall be defined as devices from a single wafer lot (for esch die type used in the construction). The conformance inspection sample shall be selected from the part category with the lowest ΔV_Z rating in the inspection lot.

Subgroup	Method	Conditions
B3	1071	Test condition E.
B4	1037	I_{ZM} = 70 mA dc at T_A = +25°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No forced air cooling on the device shall be permitted.
B5	1027	I_{ZM} = 70 mA dc for 96 hours. T_A = +75°C or adjusted as required, to give an average lot T_J = +200°C.
B6		Not applicable

4.4.2.2 Group B inspection, appendix E, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1027	Conditions for " A-1, and AUR-1" versions of: 1N4565 through 1N4569: I_Z = 0.5 mA dc, T_A = +100°C. 1N4570 through 1N4574: I_Z = 1.0 mA dc, T_A = +100°C. 1N4575 through 1N4579: I_Z = 2.0 mA dc, T_A = +100°C. 1N4580 through 1N4584: I_Z = 4.0 mA dc, T_A = +100°C.
B4, B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500.

Subgroup	Method	Conditions
C2	2036	Lead fatigue: Test condition E; tension: test condition A; 4 pounds weight, t = 15 ±3 seconds; (lead fatigue and tension tests are not applicable to surface mount "UR" version devices).
C5		Not applicable
C6	1026	Conditions for " A-1, and AUR-1" versions of: 1N4565 through 1N4569: I_Z = 0.5 mA dc, T_A = +100°C. 1N4570 through 1N4574: I_Z = 1.0 mA dc, T_A = +100°C. 1N4575 through 1N4579: I_Z = 2.0 mA dc, T_A = +100°C. 1N4580 through 1N4584: I_Z = 4.0 mA dc, T_A = +100°C.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with appendix E, table VIII of MIL-PRF-19500 and. MIL-PRF-19500, appendix G is not required when homogeneous product of radiation hard assembly (in accordance with the applicable 36D form) is submitted and successfully passes group D sampling of total dose at an outside facility compliant to MIL-STD-750, method 1019. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the submitted 36D forms.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table III herein.

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4.4.5.1 Group E inspection, appendix E, table IX of MIL-PRF-19500.

Subgroup	Method	Conditions	Qualification conformance inspection
E1	1051	500 cycles	22 devices, c = 0.
E2	1038	Condition A, 1,000 hours	22 devices, c = 0.
E3, E4 E5, E6		Not applicable	

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Voltage temperature stability. The breakdown voltage of each type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z per diode type.

4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing, at 340 hours, and at the conclusion of the life test. The 340 hour reading shall be compared with the 0-hour reading and the 1,000 hour reading compared with the 340 hour reading. The change in breakdown voltage shall not exceed the limits specified. The test temperature for breakdown voltage shall be the same as the specified ambient life test temperature.

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 s maximum) prior to reading the reference voltage. For this test, the diode shall be suspended by its' leads with mounting clips whose inside edge is located at 0.375 inch (9.53 mm) from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provide thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reference voltage	4022	(See 4.5.3)	V_Z	6.08	6.72	V dc
1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc				
1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc				
1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc				
1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc				
Small signal breakdown impedance	4051		Z_Z			ohms
1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc I_{sig} = 0.05 mA rms			200	
1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc I_{sig} = 0.1 mA rms			100	
1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc I_{sig} = 0.2 mA rms			50	
1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc I_{sig} = 0.4 mA rms			25	
Reverse current leakage	4016	DC method; V_R = 3.0 V dc	I_R		2.0	µA
Subgroup 3						
Voltage-temperature stability (see 4.5.1 and 4.5.3)		T_A = -55°C, 0°C, +25°C, +75°C, +100°C ±2°C, I_Z = (see 1.4)	ΔV_Z			mV dc
1N4565A-1, 1N4565AUR-1					100	
1N4570A-1, 1N4570AUR-1					100	
1N4575A-1, 1N4575AUR-1					100	
1N4580A-1, 1N4580AUR-1					100	
1N4566A-1, 1N4566AUR-1					50	
1N4571A-1, 1N4571AUR-1					50	
1N4576A-1, 1N4576AUR-1					50	
1N4581A-1, 1N4581AUR-1					50	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						mV dc
1N4567A-1, 1N4567AUR-1					20	
1N4572A-1, 1N4572AUR-1					20	
1N4577A-1, 1N4577AUR-1					20	
1N4582A-1, 1N4582AUR-1					20	
1N4568A-1, 1N4568AUR-1					10	
1N4573A-1, 1N4573AUR-1					10	
1N4578A-1, 1N4578AUR-1					10	
1N4583A-1, 1N4583AUR-1					10	
1N4569A-1, 1N4569AUR-1					5	
1N4574A-1, 1N4574AUR-1					5	
1N4579A-1, 1N4579AUR-1					5	
1N4584A-1, 1N4584AUR-1					5	
Subgroups 4, 5, and 6						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		$\Delta V_{Z\theta}$ Pre-Post I_{RRAD} change		Unit
	Method	Conditions		M, D, R, and H		M, D, R, and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = (see 1.4) condition A								
Reference voltage (see 4.5.3)	4022	I_Z = (see 1.4)	V_Z	6.08	6.72	6.08	6.72			V dc
Small-signal breakdown impedance	4051	I_Z = (see 1.4)	Z_Z							Ω
		I_{sig} = 0.05 ac			200		200			
1N4565A		I_{sig} = 0.05 ac			200		200			
1N4566A		I_{sig} = 0.05 ac			200		200			
1N4567A		I_{sig} = 0.05 ac			200		200			
1N4568A		I_{sig} = 0.05 ac			200		200			
1N4569A										
1N4570A		I_{sig} = 0.1 ac			100		100			
1N4571A		I_{sig} = 0.1 ac			100		100			
1N4572A		I_{sig} = 0.1 ac			100		100			
1N4573A		I_{sig} = 0.1 ac			100		100			
1N4574A		I_{sig} = 0.1 ac			100		100			
1N4575A		I_{sig} = 0.2 ac			50		50			
1N4576A		I_{sig} = 0.2 ac			50		50			
1N4577A		I_{sig} = 0.2 ac			50		50			
1N4578A		I_{sig} = 0.2 ac			50		50			
1N4579A		I_{sig} = 0.2 ac			50		50			
1N4580A		I_{sig} = 0.4 ac			25		25			
1N4581A		I_{sig} = 0.4 ac			25		25			
1N4582A		I_{sig} = 0.4 ac			25		25			
1N4583A		I_{sig} = 0.4 ac			25		25			
1N4584A		I_{sig} = 0.4 ac			25		25			

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		$\Delta V_{Z\theta}$ Pre-Post I_{RRAD} change		Unit
	Method	Conditions		M, D, R, and H		M, D, R, and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 2 - Cont										
Reverse current leakage	4016	DC method; V_R = 3.0 V dc	I_R		2.0		2.0			µA
Voltage stability (see 4.5.3)	4022	I_Z = (see 1.4) T_A = +25°C ±2°C	ΔV_Z							mV
1N4565A									± 3	
1N4566A									± 3	
1N4567A									± 2	
1N4568A									±1.5	
1N4569A									± 1	
1N4570A									± 3	
1N4571A									± 3	
1N4572A									± 2	
1N4573A									±1.5	
1N4574A									± 1	
1N4575A									± 3	
1N4576A									± 3	
1N4577A									± 2	
1N4578A									±1.5	
1N4579A									± 1	
1N4580A									± 3	
1N4581A									± 3	
1N4582A									± 2	
1N4583A									±1.5	
1N4584A									± 1	

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics and test conditions for " A-1, and AUR-1" devices are identical.

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TABLE III. Groups B and C electrical measurements. 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions 1/		Min	Max	
1.	Reference voltage	4022	(See 4.5.3)	V_Z	6.08	6.72	V dc
	1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc				
	1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc				
	1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc				
	1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc				
2.	Small-signal breakdown impedance	4051		Z_Z			Ω
	1N4565 through 1N4569		I_Z = 0.5 ±0.01 mA dc I_{sig} = 0.05 mA rms			200	
	1N4570 through 1N4574		I_Z = 1.0 ±0.01 mA dc I_{sig} = 0.1 mA rms			100	
	1N4575 through 1N4579		I_Z = 2.0 ±0.01 mA dc I_{sig} = 0.2 mA rms			50	
	1N4580 through 1N4584		I_Z = 4.0 ±0.01 mA dc I_{sig} = 0.4 mA rms			25	
3.	Reverse current leakage	4016	V_R = 3.0 V dc; DC method	I_R		2.0	µA
4.	Voltage-temperature stability (see 4.5.1 and 4.5.3)		T_A = -55°C, +0°C +75°C, +25°C,+100°C ±2° C	ΔV_Z			mV dc
	1N4565, 1N4570					100	
	1N4575, 1N4580					100	
	1N4566, 1N4571					50	
	1N4576, 1N4581					50	
	1N4567, 1N4572					20	
	1N4577, 1N4582					20	
	1N4568, 1N4573,					10	
	1N4578, 1N4583					10	
	1N4569, 1N4574					5	
	1N4579, 1N4584					5	

See footnotes at end of table.

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TABLE III. Groups B and C electrical measurements - Continued. 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Condition 1/		Min	Max	
5.	Reference-voltage stability		T_A = +100°C ±2°C (0 to 340 hours (see 4.5.1 and 4.5.3)	ΔV_Z			mV dc
	1N4565, 1N4570					10	
	1N4575, 1N4580					10	
	1N4566, 1N4571					8	
	1N4576, 1N4581					8	
	1N4567, 1N4572					7	
	1N4577, 1N4582					7	
	1N4568, 1N4573,					6	
	1N4578, 1N4583					6	
	1N4569, 1N4574					5	
	1N4579, 1N4584					5	
6.	Reference-voltage stability		T_A = +100°C ±2°C (340 to 1,000 hours (see 4.5.1 and 4.5.3)	ΔV_Z			mV dc
	1N4565, 1N4570					5	
	1N4575, 1N4580					5	
	1N4566, 1N4571					4	
	1N4576, 1N4581					4	
	1N4567, 1N4572					4	
	1N4577, 1N4582					4	
	1N4568, 1N4573,					3	
	1N4578, 1N4583					3	
	1N4569, 1N4574					3	
	1N4579, 1N4584					3	

1/ Electrical characteristics and test conditions for " A-1, and AUR-1" devices are identical.

|| 2/The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 3, see table III herein, steps 1, 2 and 3.

b. Subgroups 4 and 5, see table III herein, steps 1, 2, and 3.

3/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 2, see table III herein, steps 1, 2 and 3.

b. Subgroups 3 and 6, see table III herein, steps 1, 2 and 3.

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TABLE III. Groups B and C electrical measurements - Continued. 2/ 3/ 4/ 5/

4/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

a. Subgroup 2 see table III herein, steps 1, 2 and 3.

b. Subgroup 3, see table III herein, steps 1, 2 and 3.

c. Subgroup 6, (0 to 340 hours) see table III herein, step 5; (434 to 1,000 hours) see table III herein, steps 1, 2, 3, 5, and 6.

5/ The electrical measurements for appendix E, table VIII of MIL-PRF-19500 are as follows:

a. Subgroup 1 see table III herein, steps 1, 2 and 3.

b. Subgroup 2 see table III herein, steps 1, 2 and 3.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCAM1N4565A) will be identified on the parts list of MIL-PRF-19500.

JANC ordering information					
PIN	Manufacturer				
	55801	55801 Radiation designators M,D,L,R,F,G,H	12954	12954 Radiation designators M,D,L,R,F,G,H	12954
1N4565A through 1N4584A	JANHCA1N4565A through JANHCA1N4584A	JANHCA1N4565A through JANHCA1N4584A	JANHCB1N4565A through JANHCB1N4584A	JANHCB1N4565A through JANHCB1N4584A	JANHCC1N4565A through JANHCC1N4584A
1N4565A through 1N4584A	JANKCA1N4565A through JANKCA1N4584A	JANKCA1N4565A through JANKCA1N4584A	JANKCB1N4565A through JANKCB1N4584A	JANKCB1N4565A through JANKCB1N4584A	JANKCC1N4565A through JANKCC1N4584A

6.5 Substitution of radiation hardened devices. See MIL-PRF-19500.

6.6 Substitution of devices. ΔV_Z Device types within this series with higher type numbers (lower ΔV_Z) are a direct oneway substitution for lower type numbers (higher ΔV_Z).

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1662)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-PRF-19500/452E**	**2. DOCUMENT DATE (YYMMDD)** 971001

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, LOW LEVEL VOLTAGE-REFERENCE, TYPES 1N4565A-1, THROUGH 1N4584A-1, 1N4565AUR-1, THROUGH 1N4584AUR-1, JAN, JANTX, JANTXV, JANS, JANHC AND JANKC; RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H; JANSM, D, L, R, F, G, H; JANHCM, D, L, R, F, G, H; JANKCM, D, L, R, F, G, H

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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